

GOAD - part 1 - reconnaissance and scan

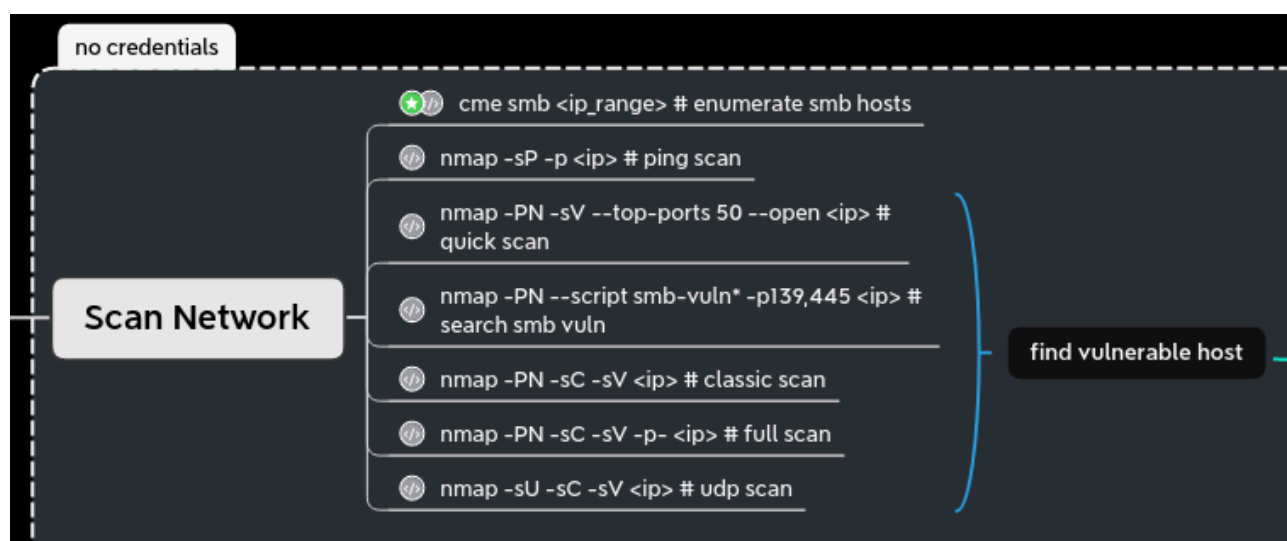
mayfly277.github.io/posts/GOADv2-pwning_part1

July 2, 2022

The lab is now up and running [Goad introduction](#), let's do some recon on it.

Enumerate Network

We will starting the reconnaissance of the Game Of Active Directory environment by searching all the availables IPs.



First recon with cme

The first thing i personally do before launching an nmap is to scan for netbios results. For this i launch crackmapexec (cme) on the IP range to quickly get netbios answers by windows computers. This is a very kick way to get all the windows machine IP, names and domains.

```
cme smb  
192.168.56.1/24
```

```
[Jul 03, 2022 - 16:19:14 (CEST)] exegol-goadv2 /workspace # cme smb 192.168.56.0/24  
SMB 192.168.56.23 445 BRAAVOS [*] Windows Server 2016 Standard Evaluation 14393 x64 (name:BRAAVOS) (domain:essos.local) (signing:False) (SMBv1:True)  
SMB 192.168.56.11 445 WINTERFELL [*] Windows 10.0 Build 17763 x64 (name:WINTERFELL) (domain:north.sevenkingdoms.local) (signing:True) (SMBv1:False)  
SMB 192.168.56.12 445 MEEREEN [*] Windows Server 2016 Standard Evaluation 14393 x64 (name:MEEREEN) (domain:essos.local) (signing:True) (SMBv1:True)  
SMB 192.168.56.22 445 CASTELBLACK [*] Windows 10.0 Build 17763 x64 (name:CASTELBLACK) (domain:north.sevenkingdoms.local) (signing:False) (SMBv1:False)  
SMB 192.168.56.10 445 KINGSLANDING [*] Windows 10.0 Build 17763 x64 (name:KINGSLANDING) (domain:sevenkingdoms.local) (signing:True) (SMBv1:False)
```

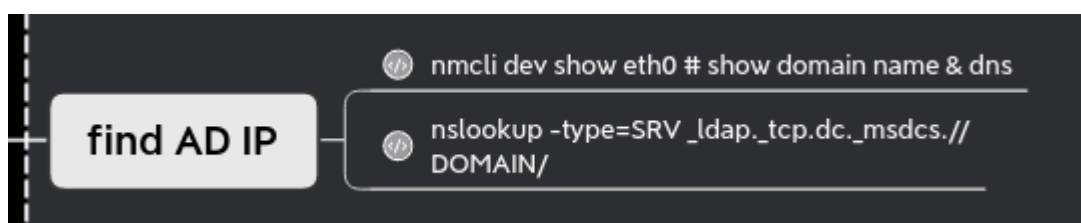
This command answer relatively quickly and send back a lot of useful informations!

We now know there is 3 domains:

- north.sevenkingdoms.local (2 ip)
 - CASTELBLACK (windows server 2019) (signing false)
 - WINTERFELL (windows server 2019)
- sevenkingdoms.local (1 ip)
 - KINGSLANDING (windows server 2019)
- essos.local (2 ip)
 - BRAAVOS (windows server 2016) (signing false)
 - MEEREEN (windows server 2019)

Here as we have 3 domains we know that three DCs must be setup. We also know that microsoft setup DC smb signing as true by default. So all the dc are the one with signing at true. (In a secure environment signing must true everywhere to avoid ntlm relay).

Find DC ip



Let's enumerate the DCs by querying the dns with nslookup

```
nslookup -type=srv _ldap._tcp.dc._msdcs.sevenkingdoms.local
192.168.56.10
```

```
[Jul 04, 2022 - 23:02:53 (CEST)] exegol-goadv2 /workspace # nslookup -type=srv _ldap._tcp.dc._msdcs.sevenkingdoms.local 192.168.56.10
Server:
192.168.56.10
Address:
192.168.56.10#53

_ldap._tcp.dc._msdcs.sevenkingdoms.local      service = 0 100 389 kingslanding.sevenkingdoms.local.

[Jul 04, 2022 - 23:03:01 (CEST)] exegol-goadv2 /workspace # nslookup -type=srv _ldap._tcp.dc._msdcs.north.sevenkingdoms.local 192.168.56.10
Server:
192.168.56.10
Address:
192.168.56.10#53

Non-authoritative answer:
_ldap._tcp.dc._msdcs.north.sevenkingdoms.local service = 0 100 389 winterfell.north.sevenkingdoms.local.

Authoritative answers can be found from:
winterfell.north.sevenkingdoms.local    internet address = 192.168.56.11

[Jul 04, 2022 - 23:05:36 (CEST)] exegol-goadv2 /workspace # nslookup -type=srv _ldap._tcp.dc._msdcs.essos.local 192.168.56.10
Server:
192.168.56.10
Address:
192.168.56.10#53

Non-authoritative answer:
_ldap._tcp.dc._msdcs.essos.local      service = 0 100 389 meereen.essos.local.

Authoritative answers can be found from:
meereen.essos.local    internet address = 192.168.56.12
meereen.essos.local    internet address = 10.0.2.15
```

setup /etc/hosts and kerberos

- To use kerberos in our Linux environment we will do some configurations.
- First we must set the DNS by configuring the /etc/hosts file

```
# /etc/hosts
# GOAD
192.168.56.10  sevenkingdoms.local kingslanding.sevenkingdoms.local kingslanding
192.168.56.11  winterfell.north.sevenkingdoms.local north.sevenkingdoms.local winterfell
192.168.56.12  essos.local meereen.essos.local meereen
192.168.56.22  castelblack.north.sevenkingdoms.local castelblack
192.168.56.23  braavos.essos.local braavos
```

We must install the Linux kerberos client

```
sudo apt install krb5-
user
```

- We answer the questions with :
 - realm : essos.local
 - servers : meereen.essos.local
- We will setup the /etc/krb5.conf file like this :

```
[libdefaults]
    default_realm = essos.local
    kdc_timesync = 1
    ccache_type = 4
    forwardable = true
    proxiable = true
    fcc-mit-ticketflags = true
[realms]
    north.sevenkingdoms.local = {
        kdc = winterfell.north.sevenkingdoms.local
        admin_server = winterfell.north.sevenkingdoms.local
    }
    sevenkingdoms.local = {
        kdc = kingslanding.sevenkingdoms.local
        admin_server = kingslanding.sevenkingdoms.local
    }
    essos.local = {
        kdc = meereen.essos.local
        admin_server = meereen.essos.local
    }
    ...
```

- If krb5-user is already installed we can reconfigure it with (dpkg-reconfigure or by modifying /etc/krb5.conf)

```
dpkg-reconfigure krb5-
config
```

Now kerberos is set up on our environment we will try if we can get a TGT for a user.

```
getTGT.py
essos.local/khal.drogo:horse
```

[*] Saving ticket in khal.drogo.ccache

```
export
KRB5CCNAME=/workspace/khal.drogo.ccache
```

```
smbclient.py -k
@braavos.essos.local
```

Type help for list of commands

```
# shares
ADMIN$
all
C$
CertEnroll
IPC$
public
# use C$
# ls
drw-rw-rw-      0 Wed Jun 29 10:41:11 2022 $Recycle.Bin
-rw-rw-rw- 384322 Thu Feb 14 20:38:48 2019 bootmgr
-rw-rw-rw-      1 Thu Feb 14 20:38:48 2019 BOOTNXT
drw-rw-rw-      0 Wed Jun 29 01:20:28 2022 Config.Msi
-rw-rw-rw- 1914 Wed Jun 29 07:46:13 2022 dns_log.txt
drw-rw-rw-      0 Wed Jun 29 08:29:08 2022 Documents and Settings
drw-rw-rw-      0 Wed Jun 29 01:26:54 2022 inetpub
-rw-rw-rw- 1342177280 Sun Jul  3 22:57:35 2022 pagefile.sys
drw-rw-rw-      0 Thu Feb 14 13:19:12 2019 PerfLogs
drw-rw-rw-      0 Wed Jun 29 01:19:59 2022 Program Files
drw-rw-rw-      0 Wed Jun 29 01:29:07 2022 Program Files (x86)
drw-rw-rw-      0 Wed Jun 29 10:46:28 2022 ProgramData
drw-rw-rw-      0 Wed Jun 29 00:29:20 2022 Recovery
drw-rw-rw-      0 Wed Jun 29 01:07:10 2022 setup
drw-rw-rw-      0 Wed Jun 29 01:31:47 2022 shares
drw-rw-rw-      0 Thu Feb 14 20:40:57 2019 System Volume Information
drw-rw-rw-      0 Wed Jun 29 01:01:06 2022 tmp
drw-rw-rw-      0 Wed Jun 29 10:40:42 2022 Users
drw-rw-rw-      0 Sun Jul  3 15:58:04 2022 vagrant
drw-rw-rw-      0 Thu Jun 30 17:33:12 2022 Windows
```

- Ok the kerberos setup is good :)
- We could now unset the ticket:
- Trouble on winterfell
- During the kerberos tests we saw we get trouble on winterfell:

```
getTGT.py
north.sevenkingdoms.local/arya.stark:Needle
```

Impacket v0.10.0 - Copyright 2022 SecureAuth Corporation

[*] Saving ticket in arya.stark.ccache

```
export
KRB5CCNAME=/workspace/arya.stark.ccache
```

```
smbclient.py -k -no-pass
@winterfell.north.sevenkingdoms.local
```

Impacket v0.10.0 - Copyright 2022 SecureAuth Corporation

[-] SMB SessionError: STATUS_MORE_PROCESSING_REQUIRED({Still Busy} The specified I/O request packet (IRP) cannot be disposed of because the I/O operation is not complete.)

Actually i don't know why kerberos doesn't work on winterfell with the full FQDN, but it's ok by just setting winterfell instead of winterfell.north.sevenkingdoms.local

```
smbclient.py -k -no-pass
@winterfell
```

Impacket v0.10.0 - Copyright 2022 SecureAuth Corporation

Type help for list of commands

shares

ADMIN\$

C\$

IPC\$

NETLOGON

SYSVOL

use SYSVOL

ls

```
drw-rw-rw-    0 Wed Jun 29 00:48:39 2022 .
drw-rw-rw-    0 Wed Jun 29 00:48:39 2022 ..
drw-rw-rw-    0 Wed Jun 29 00:48:39 2022 north.sevenkingdoms.local
```

Nmap

One thing to know is that nmap will do a ping before scanning the target. If the target doesn't answer to ping it will be ignore.

The way to be sure we doesn't miss anything on TCP, could be to scan with the following options:

```
nmap -Pn -p- -sC -sV -oA full_scan_goad 192.168.56.10-
12,22-23
```

Let's analyze this command :

- -Pn don't do ping scan and scan all ip

- -p- scan the 65535 ports instead of the default nmap 1000 top ports by default
- -sC play the default script for reconnaissance
- -sV enumerate the version
- -oA write results in the 3 available format (nmap classic, grep format, xml format)
- The full scan result is:

```
Nmap scan report for sevenkingdoms.local (192.168.56.10)
Host is up (0.00066s latency).
Not shown: 65511 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
53/tcp    open  domain       Simple DNS Plus
80/tcp    open  http         Microsoft IIS httpd 10.0
|_http-server-header: Microsoft-IIS/10.0
|_http-methods:
|_ Potentially risky methods: TRACE
|_http-title: IIS Windows Server
88/tcp    open  kerberos-sec Microsoft Windows Kerberos (server time: 2022-07-03 15:14:43Z)
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
389/tcp   open  ldap         Microsoft Windows Active Directory LDAP (Domain:
sevenkingdoms.local0., Site: Default-First-Site-Name)
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
|_ssl-cert: Subject: commonName=kingslanding.sevenkingdoms.local
|_ Subject Alternative Name: othername:<unsupported>, DNS:kingslanding.sevenkingdoms.local
|_ Not valid before: 2022-06-30T06:55:19
|_ Not valid after: 2023-06-30T06:55:19
445/tcp   open  microsoft-ds?
464/tcp   open  kpasswd5?
593/tcp   open  ncacn_http   Microsoft Windows RPC over HTTP 1.0
636/tcp   open  ssl/ldap     Microsoft Windows Active Directory LDAP (Domain:
sevenkingdoms.local0., Site: Default-First-Site-Name)
|_ssl-cert: Subject: commonName=kingslanding.sevenkingdoms.local
|_ Subject Alternative Name: othername:<unsupported>, DNS:kingslanding.sevenkingdoms.local
|_ Not valid before: 2022-06-30T06:55:19
|_ Not valid after: 2023-06-30T06:55:19
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
3268/tcp  open  ldap         Microsoft Windows Active Directory LDAP (Domain:
sevenkingdoms.local0., Site: Default-First-Site-Name)
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
|_ssl-cert: Subject: commonName=kingslanding.sevenkingdoms.local
|_ Subject Alternative Name: othername:<unsupported>, DNS:kingslanding.sevenkingdoms.local
|_ Not valid before: 2022-06-30T06:55:19
|_ Not valid after: 2023-06-30T06:55:19
3269/tcp  open  ssl/ldap     Microsoft Windows Active Directory LDAP (Domain:
sevenkingdoms.local0., Site: Default-First-Site-Name)
|_ssl-cert: Subject: commonName=kingslanding.sevenkingdoms.local
|_ Subject Alternative Name: othername:<unsupported>, DNS:kingslanding.sevenkingdoms.local
|_ Not valid before: 2022-06-30T06:55:19
|_ Not valid after: 2023-06-30T06:55:19
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
|_ssl-cert: Subject: commonName=kingslanding.sevenkingdoms.local
|_ Not valid before: 2022-06-27T22:40:05
|_ Not valid after: 2022-12-27T22:40:05
|_rdp-ntlm-info:
|_ Target_Name: SEVENKINGDOMS
```

```
| NetBIOS_Domain_Name: SEVENKINGDOMS
| NetBIOS_Computer_Name: KINGSLANDING
| DNS_Domain_Name: sevenkingdoms.local
| DNS_Computer_Name: kingslanding.sevenkingdoms.local
| DNS_Tree_Name: sevenkingdoms.local
| Product_Version: 10.0.17763
|_ System_Time: 2022-07-03T15:16:37+00:00
5985/tcp open http Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Not Found
5986/tcp open ssl/http Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
|_tls-alpn:
|_ http/1.1
|_ssl-cert: Subject: commonName=VAGRANT
| Subject Alternative Name: DNS:VAGRANT, DNS:vagrant
| Not valid before: 2022-06-27T15:18:01
|_Not valid after: 2025-06-26T15:18:01
|_http-title: Not Found
9389/tcp open mc-nmf .NET Message Framing
49666/tcp open msrpc Microsoft Windows RPC
49667/tcp open msrpc Microsoft Windows RPC
49677/tcp open ncacn_http Microsoft Windows RPC over HTTP 1.0
49678/tcp open msrpc Microsoft Windows RPC
49680/tcp open msrpc Microsoft Windows RPC
49693/tcp open msrpc Microsoft Windows RPC
49708/tcp open msrpc Microsoft Windows RPC
59098/tcp open msrpc Microsoft Windows RPC
MAC Address: 08:00:27:57:A4:F2 (Oracle VirtualBox virtual NIC)
Service Info: Host: KINGSLANDING; OS: Windows; CPE: cpe:/o:microsoft:windows
```

Host script results:

```
|_nbstat: NetBIOS name: KINGSLANDING, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:57:a4:f2
(Oracle VirtualBox virtual NIC)
|_smb2-time:
| date: 2022-07-03T15:16:34
|_start_date: N/A
|_smb2-security-mode:
| 3.1.1:
|_ Message signing enabled and required
```

Nmap scan report for winterfell.north.sevenkingdoms.local (192.168.56.11)

Host is up (0.00068s latency).

Not shown: 65513 filtered tcp ports (no-response)

PORT	STATE	SERVICE	VERSION
53/tcp	open	domain	Simple DNS Plus
88/tcp	open	kerberos-sec	Microsoft Windows Kerberos (server time: 2022-07-03 15:14:49Z)
135/tcp	open	msrpc	Microsoft Windows RPC
139/tcp	open	netbios-ssn	Microsoft Windows netbios-ssn
389/tcp	open	ldap	Microsoft Windows Active Directory LDAP (Domain: sevenkingdoms.local0., Site: Default-First-Site-Name)

```
|_ssl-cert: Subject: commonName=winterfell.north.sevenkingdoms.local
| Subject Alternative Name: othername:<unsupported>, DNS:winterfell.north.sevenkingdoms.local
| Not valid before: 2022-06-30T13:20:52
|_Not valid after: 2023-06-30T13:20:52
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
445/tcp open microsoft-ds?
464/tcp open kpasswd?
593/tcp open ncacn_http Microsoft Windows RPC over HTTP 1.0
```

```

636/tcp open  ssl/ldap      Microsoft Windows Active Directory LDAP (Domain:
sevenkingdoms.local0., Site: Default-First-Site-Name)
| ssl-cert: Subject: commonName=winterfell.north.sevenkingdoms.local
| Subject Alternative Name: othername:<unsupported>, DNS:winterfell.north.sevenkingdoms.local
| Not valid before: 2022-06-30T13:20:52
|_Not valid after: 2023-06-30T13:20:52
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
3268/tcp open  ldap          Microsoft Windows Active Directory LDAP (Domain:
sevenkingdoms.local0., Site: Default-First-Site-Name)
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
| ssl-cert: Subject: commonName=winterfell.north.sevenkingdoms.local
| Subject Alternative Name: othername:<unsupported>, DNS:winterfell.north.sevenkingdoms.local
| Not valid before: 2022-06-30T13:20:52
|_Not valid after: 2023-06-30T13:20:52
3269/tcp open  ssl/ldap      Microsoft Windows Active Directory LDAP (Domain:
sevenkingdoms.local0., Site: Default-First-Site-Name)
| ssl-cert: Subject: commonName=winterfell.north.sevenkingdoms.local
| Subject Alternative Name: othername:<unsupported>, DNS:winterfell.north.sevenkingdoms.local
| Not valid before: 2022-06-30T13:20:52
|_Not valid after: 2023-06-30T13:20:52
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
3389/tcp open  ms-wbt-server Microsoft Terminal Services
| ssl-cert: Subject: commonName=winterfell.north.sevenkingdoms.local
| Not valid before: 2022-06-27T22:49:00
|_Not valid after: 2022-12-27T22:49:00
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
| rdp-ntlm-info:
|   Target_Name: NORTH
|   NetBIOS_Domain_Name: NORTH
|   NetBIOS_Computer_Name: WINTERFELL
|   DNS_Domain_Name: north.sevenkingdoms.local
|   DNS_Computer_Name: winterfell.north.sevenkingdoms.local
|   DNS_Tree_Name: sevenkingdoms.local
|   Product_Version: 10.0.17763
|_ System_Time: 2022-07-03T15:16:36+00:00
5985/tcp open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-title: Not Found
|_http-server-header: Microsoft-HTTPAPI/2.0
5986/tcp open  ssl/http      Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
| tls-alpn:
|_ http/1.1
|_http-server-header: Microsoft-HTTPAPI/2.0
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
|_http-title: Not Found
| ssl-cert: Subject: commonName=VAGRANT
| Subject Alternative Name: DNS:VAGRANT, DNS:vagrant
| Not valid before: 2022-06-27T15:21:08
|_Not valid after: 2025-06-26T15:21:08
9389/tcp open  mc-nmf        .NET Message Framing
49666/tcp open  msrpc         Microsoft Windows RPC
49667/tcp open  msrpc         Microsoft Windows RPC
49669/tcp open  ncacn_http    Microsoft Windows RPC over HTTP 1.0
49670/tcp open  msrpc         Microsoft Windows RPC
49672/tcp open  msrpc         Microsoft Windows RPC
49702/tcp open  msrpc         Microsoft Windows RPC
65091/tcp open  msrpc         Microsoft Windows RPC
MAC Address: 08:00:27:3E:C3:EF (Oracle VirtualBox virtual NIC)
Service Info: Host: WINTERFELL; OS: Windows; CPE: cpe:/o:microsoft:windows

```

Host script results:


```
|_nbstat: NetBIOS name: WINTERFELL, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:3e:c3:ef
(Oracle VirtualBox virtual NIC)
| smb2-time:
|   date: 2022-07-03T15:16:37
|_ start_date: N/A
| smb2-security-mode:
|   3.1.1:
|_   Message signing enabled and required
```

Nmap scan report for essos.local (192.168.56.12)

Host is up (0.00046s latency).

Not shown: 65513 filtered tcp ports (no-response)

PORT	STATE	SERVICE	VERSION
53/tcp	open	domain	Simple DNS Plus
88/tcp	open	kerberos-sec	Microsoft Windows Kerberos (server time: 2022-07-03 15:15:01Z)
135/tcp	open	msrpc	Microsoft Windows RPC
139/tcp	open	netbios-ssn	Microsoft Windows netbios-ssn
389/tcp	open	ldap	Microsoft Windows Active Directory LDAP (Domain: essos.local, Site: Default-First-Site-Name)

```
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
| ssl-cert: Subject: commonName=meereen.essos.local
| Subject Alternative Name: othername:<unsupported>, DNS:meereen.essos.local
| Not valid before: 2022-06-29T08:30:39
|_Not valid after: 2023-06-29T08:30:39
445/tcp open microsoft-ds Windows Server 2016 Standard Evaluation 14393 microsoft-ds
(workgroup: ESSOS)
464/tcp open kpasswd5?
593/tcp open ncacn_http Microsoft Windows RPC over HTTP 1.0
636/tcp open ssl/ldap Microsoft Windows Active Directory LDAP (Domain: essos.local,
Site: Default-First-Site-Name)
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
| ssl-cert: Subject: commonName=meereen.essos.local
| Subject Alternative Name: othername:<unsupported>, DNS:meereen.essos.local
| Not valid before: 2022-06-29T08:30:39
|_Not valid after: 2023-06-29T08:30:39
3268/tcp open ldap Microsoft Windows Active Directory LDAP (Domain: essos.local,
Site: Default-First-Site-Name)
| ssl-cert: Subject: commonName=meereen.essos.local
| Subject Alternative Name: othername:<unsupported>, DNS:meereen.essos.local
| Not valid before: 2022-06-29T08:30:39
|_Not valid after: 2023-06-29T08:30:39
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
3269/tcp open ssl/ldap Microsoft Windows Active Directory LDAP (Domain: essos.local,
Site: Default-First-Site-Name)
| ssl-cert: Subject: commonName=meereen.essos.local
| Subject Alternative Name: othername:<unsupported>, DNS:meereen.essos.local
| Not valid before: 2022-06-29T08:30:39
|_Not valid after: 2023-06-29T08:30:39
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
3389/tcp open ms-wbt-server Microsoft Terminal Services
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
| ssl-cert: Subject: commonName=meereen.essos.local
| Not valid before: 2022-06-27T22:40:08
|_Not valid after: 2022-12-27T22:40:08
| rdp-ntlm-info:
|   Target_Name: ESSOS
|   NetBIOS_Domain_Name: ESSOS
|   NetBIOS_Computer_Name: MEEREEN
|   DNS_Domain_Name: essos.local
|   DNS_Computer_Name: meereen.essos.local
```

```

|   DNS_Tree_Name: essos.local
|   Product_Version: 10.0.14393
|_  System_Time: 2022-07-03T15:16:38+00:00
5985/tcp open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Not Found
5986/tcp open  ssl/http        Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
|_http-title: Not Found
|  ssl-cert: Subject: commonName=VAGRANT
|  Subject Alternative Name: DNS:VAGRANT, DNS:vagrant
|  Not valid before: 2022-06-27T15:23:42
|_Not valid after: 2025-06-26T15:23:42
|_http-server-header: Microsoft-HTTPAPI/2.0
|  tls-alpn:
|    h2
|_  http/1.1
9389/tcp open  mc-nmf          .NET Message Framing
49666/tcp open  msrpc          Microsoft Windows RPC
49667/tcp open  msrpc          Microsoft Windows RPC
49677/tcp open  ncacn_http     Microsoft Windows RPC over HTTP 1.0
49678/tcp open  msrpc          Microsoft Windows RPC
49680/tcp open  msrpc          Microsoft Windows RPC
49693/tcp open  msrpc          Microsoft Windows RPC
60502/tcp open  msrpc          Microsoft Windows RPC
MAC Address: 08:00:27:33:DF:2F (Oracle VirtualBox virtual NIC)
Service Info: Host: MEEREEN; OS: Windows; CPE: cpe:/o:microsoft:windows

```

Host script results:

```

| smb2-time:
|   date: 2022-07-03T15:16:38
|_  start_date: 2022-07-03T13:56:01
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled and required
|_nbstat: NetBIOS name: MEEREEN, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:33:df:2f
(Oracle VirtualBox virtual NIC)
|_clock-skew: mean: 42m00s, deviation: 2h12m50s, median: 0s
| smb-security-mode:
|   account_used: <blank>
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: required
| smb-os-discovery:
|   OS: Windows Server 2016 Standard Evaluation 14393 (Windows Server 2016 Standard Evaluation
6.3)
|   Computer name: meereen
|   NetBIOS computer name: MEEREEN\x00
|   Domain name: essos.local
|   Forest name: essos.local
|   FQDN: meereen.essos.local
|_  System time: 2022-07-03T08:16:38-07:00

```

Nmap scan report for castelblack.north.sevenkingdoms.local (192.168.56.22)

Host is up (0.00066s latency).

Not shown: 65525 filtered tcp ports (no-response)

```

PORT      STATE SERVICE      VERSION
80/tcp    open  http          Microsoft IIS httpd 10.0
|_http-title: Site doesn't have a title (text/html).
| http-methods:

```

```

|_ Potentially risky methods: TRACE
|_http-server-header: Microsoft-IIS/10.0
135/tcp open msrpc Microsoft Windows RPC
139/tcp open netbios-ssn Microsoft Windows netbios-ssn
445/tcp open microsoft-ds?
1433/tcp open ms-sql-s Microsoft SQL Server 2019 15.00.2000.00; RTM
| ssl-cert: Subject: commonName=SSL_Self_Signed_Fallback
| Not valid before: 2022-07-03T13:57:05
|_Not valid after: 2052-07-03T13:57:05
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
| ms-sql-ntlm-info:
|   Target_Name: NORTH
|   NetBIOS_Domain_Name: NORTH
|   NetBIOS_Computer_Name: CASTELBLACK
|   DNS_Domain_Name: north.sevenkingdoms.local
|   DNS_Computer_Name: castelblack.north.sevenkingdoms.local
|   DNS_Tree_Name: sevenkingdoms.local
|_ Product_Version: 10.0.17763
3389/tcp open ms-wbt-server Microsoft Terminal Services
| ssl-cert: Subject: commonName=castelblack.north.sevenkingdoms.local
| Not valid before: 2022-06-27T22:56:08
|_Not valid after: 2022-12-27T22:56:08
| rdp-ntlm-info:
|   Target_Name: NORTH
|   NetBIOS_Domain_Name: NORTH
|   NetBIOS_Computer_Name: CASTELBLACK
|   DNS_Domain_Name: north.sevenkingdoms.local
|   DNS_Computer_Name: castelblack.north.sevenkingdoms.local
|   DNS_Tree_Name: sevenkingdoms.local
|   Product_Version: 10.0.17763
|_ System_Time: 2022-07-03T15:16:40+00:00
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
5985/tcp open http Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Not Found
5986/tcp open ssl/http Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
| ssl-cert: Subject: commonName=VAGRANT
| Subject Alternative Name: DNS:VAGRANT, DNS:vagrant
| Not valid before: 2022-06-27T15:26:52
|_Not valid after: 2025-06-26T15:26:52
|_tls-alpn:
|_ http/1.1
|_http-title: Not Found
|_http-server-header: Microsoft-HTTPAPI/2.0
|_ssl-date: 2022-07-03T15:17:20+00:00; 0s from scanner time.
49666/tcp open msrpc Microsoft Windows RPC
49677/tcp open msrpc Microsoft Windows RPC
MAC Address: 08:00:27:B9:6C:19 (Oracle VirtualBox virtual NIC)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

```

Host script results:

```

| smb2-time:
|   date: 2022-07-03T15:16:38
|_ start_date: N/A
| ms-sql-info:
|   192.168.56.22:1433:
|     Version:
|       name: Microsoft SQL Server 2019 RTM
|       number: 15.00.2000.00
|       Product: Microsoft SQL Server 2019

```

```
|      Service pack level: RTM
|      Post-SP patches applied: false
|_    TCP port: 1433
|_nbstat: NetBIOS name: CASTELBLACK, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:b9:6c:19
(Oracle VirtualBox virtual NIC)
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled but not required
```

Stats: 0:09:09 elapsed; 4 hosts completed (5 up), 1 undergoing Script Scan

NSE Timing: About 99.93% done; ETC: 17:20 (0:00:00 remaining)

Nmap scan report for braavos.essos.local (192.168.56.23)

Host is up (0.00044s latency).

Not shown: 65524 filtered tcp ports (no-response)

PORT	STATE	SERVICE	VERSION
------	-------	---------	---------

80/tcp	open	http	Microsoft IIS httpd 10.0
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|_http-server-header: Microsoft-IIS/10.0

| http-methods:

|_ Potentially risky methods: TRACE

|_http-title: IIS Windows Server

135/tcp	open	msrpc	Microsoft Windows RPC
---------	------	-------	-----------------------

139/tcp	open	netbios-ssn	Microsoft Windows netbios-ssn
---------	------	-------------	-------------------------------

445/tcp	open	microsoft-ds	Windows Server 2016 Standard Evaluation 14393 microsoft-ds
---------	------	--------------	--

1433/tcp	open	ms-sql-s	Microsoft SQL Server 2019 15.00.2000.00; RTM
----------	------	----------	--

| ms-sql-ntlm-info:

| Target_Name: ESSOS

| NetBIOS_Domain_Name: ESSOS

| NetBIOS_Computer_Name: BRAAVOS

| DNS_Domain_Name: essos.local

| DNS_Computer_Name: braavos.essos.local

| DNS_Tree_Name: essos.local

|_ Product_Version: 10.0.14393

| ssl-cert: Subject: commonName=SSL_Self_Signed_Fallback

| Not valid before: 2022-07-03T13:57:51

|_Not valid after: 2052-07-03T13:57:51

|_ssl-date: 2022-07-03T15:20:40+00:00; 0s from scanner time.

3389/tcp	open	ms-wbt-server	Microsoft Terminal Services
----------	------	---------------	-----------------------------

| rdp-ntlm-info:

| Target_Name: ESSOS

| NetBIOS_Domain_Name: ESSOS

| NetBIOS_Computer_Name: BRAAVOS

| DNS_Domain_Name: essos.local

| DNS_Computer_Name: braavos.essos.local

| DNS_Tree_Name: essos.local

| Product_Version: 10.0.14393

|_ System_Time: 2022-07-03T15:20:00+00:00

| ssl-cert: Subject: commonName=braavos.essos.local

| Not valid before: 2022-06-27T22:56:08

|_Not valid after: 2022-12-27T22:56:08

|_ssl-date: 2022-07-03T15:20:40+00:00; 0s from scanner time.

5985/tcp	open	http	Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
----------	------	------	---

|_http-server-header: Microsoft-HTTPAPI/2.0

|_http-title: Not Found

5986/tcp	open	ssl/http	Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
----------	------	----------	---

| ssl-cert: Subject: commonName=VAGRANT

| Subject Alternative Name: DNS:VAGRANT, DNS:vagrant

| Not valid before: 2022-06-27T15:30:05

|_Not valid after: 2025-06-26T15:30:05

| tls-alpn:

| h2

```

|_ http/1.1
|_ssl-date: 2022-07-03T15:20:40+00:00; 0s from scanner time.
|_http-title: Not Found
|_http-server-header: Microsoft-HTTPAPI/2.0
49669/tcp open  msrpc          Microsoft Windows RPC
49685/tcp open  msrpc          Microsoft Windows RPC
49778/tcp open  msrpc          Microsoft Windows RPC
MAC Address: 08:00:27:A3:67:1D (Oracle VirtualBox virtual NIC)
Service Info: OSs: Windows, Windows Server 2008 R2 - 2012; CPE: cpe:/o:microsoft:windows

Host script results:
| smb2-time:
|   date: 2022-07-03T15:20:00
|_ start_date: 2022-07-03T13:57:42
|_nbstat: NetBIOS name: BRAAVOS, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:a3:67:1d
(Oracle VirtualBox virtual NIC)
| ms-sql-info:
|   192.168.56.23:1433:
|     Version:
|       name: Microsoft SQL Server 2019 RTM
|       number: 15.00.2000.00
|       Product: Microsoft SQL Server 2019
|       Service pack level: RTM
|       Post-SP patches applied: false
|_   TCP port: 1433
| smb2-security-mode:
|   3.1.1:
|_   Message signing enabled but not required
| smb-os-discovery:
|   OS: Windows Server 2016 Standard Evaluation 14393 (Windows Server 2016 Standard Evaluation
6.3)
|   Computer name: braavos
|   NetBIOS computer name: BRAAVOS\x00
|   Domain name: essos.local
|   Forest name: essos.local
|   FQDN: braavos.essos.local
|_   System time: 2022-07-03T08:20:00-07:00
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_   message_signing: disabled (dangerous, but default)
|_clock-skew: mean: 52m29s, deviation: 2h28m29s, median: 0s

Post-scan script results:
| clock-skew:
|   0s:
|     192.168.56.11 (winterfell.north.sevenkingdoms.local)
|     192.168.56.22 (castelblack.north.sevenkingdoms.local)
|     192.168.56.12 (essos.local)
|     192.168.56.10 (sevenkingdoms.local)
|_   192.168.56.23 (braavos.essos.local)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 5 IP addresses (5 hosts up) scanned in 572.00 seconds

```

Ok we now know all the hosts and service exposed, let try an anonymous enumeration in the second part: [Goad pwning part2](#)